Back **Week Three Quiz**
Graded Assignment

Due Mar 16, 11:59 PM CET

1. When a variable has rank orders, it is a: 1 point

- nominal categorical variable
- ordinal categorical variable
- measurable variable

2. Only proportionate transformations of the form $y = bx$ are permitted for: 1 point

- interval measurable variables
- ratio measurable variables

3. Statistical inference: 1 point

- summarises the data which were collected, in order to make them more understandable
- uses the observed data to draw conclusions about some broader population

4. A scatterplot is ideal for showing the values of: 1 point

- two categorical variables
- two measurable variables

5. Variance and standard deviation are: 1 point

- measures of central tendency
- measures of spread

6. For a set of sample data: 1 point

- $\sum_{i=1}^{n}(X_i - \bar{X}) < 0$
- $\sum_{i=1}^{n}(X_i - \bar{X}) = 0$
- $\sum_{i=1}^{n}(X_i - \bar{X}) > 0$

7. For many symmetric distributions, about 2/3 of the observations are: 1 point

- between $\bar{X} - S$ and $\bar{X} + S$, that is, within one (sample) standard deviation about the (sample) mean
- between $\bar{X} - 2 \times S$ and $\bar{X} + 2 \times S$, that is, within two (sample) standard deviations about the (sample) mean

8. If $X \sim N(\mu, \sigma^2)$, and $Y = aX + b$, for $a > 0$ and $b > 0$ then: 1 point

- $Y \sim N(a\mu + b,\ a^2\sigma^2)$
- $Y \sim N(a\mu + b,\ a^2\sigma^2 + b^2)$

9. The variance of a (discrete) random variable, σ^2, is a probability-weighted average given by: 1 point

- $\sum_{i=1}^{N}(x_i - \mu)\, p(x_i)$
- $\sum_{i=1}^{N}(x_i - \mu)^2\, p(x_i)$

10. If $Z \sim N(0, 1)$, then $P(-1 \leq Z \leq 1)$: 1 point

- ≈ 0.683
- ≈ 0.950
- ≈ 0.997